

From four levels to the Dicke model

a step-by-step reduction for cavity Rydberg-EIT electrometry

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Abstract

We start from the exact laboratory Hamiltonian and reduce it, one controlled step at a time, to a collective spin–boson model. Each step is a matrix operation and each carries a stated validity condition. Two results deserve advance notice. First, the enormous $|r\rangle \rightarrow |r'\rangle$ splitting does not appear in the reduced model at all: the rotating frame absorbs it, leaving only the microwave *detuning* Δ_μ . Second, the microwave dressing generally produces two branches that couple to the ground state *unequally*, by an amount set by the mixing angle; the familiar symmetric Autler–Townes doublet is the $\Delta_\mu = 0$ limit, and the single-branch (AC Stark) picture is the $|\Delta_\mu| \gg \Omega_\mu$ limit. Which limit one works in decides whether the measurand enters linearly or quadratically. Throughout $\hbar = 1$.

1 The exact Hamiltonian

Four atomic levels with laboratory energies $\omega_1 = 0 < \omega_2 < \omega_3 < \omega_4$:

$$|1\rangle = |5S_{1/2}\rangle, \quad |2\rangle = |5P_{3/2}\rangle, \quad |3\rangle = |53D_{5/2}\rangle, \quad |4\rangle = |54P_{3/2}\rangle,$$

numerically $\omega_2/2\pi = 384.25$, $\omega_3/2\pi = 1008.82$ THz, and $(\omega_4 - \omega_3)/2\pi \simeq 10$ GHz. Writing $\sigma_{jk} = |j\rangle\langle k|$, and with no approximations whatsoever,

$$\begin{aligned} H = & \underbrace{\sum_{j=1}^4 \omega_j \sigma_{jj}}_{\text{free}} + \underbrace{g_0 (a + a^\dagger)(\sigma_{12} + \sigma_{21})}_{\text{cavity, } |1\rangle \leftrightarrow |2\rangle} \\ & + \underbrace{\Omega_c \cos(\nu_c t) (\sigma_{23} + \sigma_{32})}_{\text{control laser}} + \underbrace{\Omega_\mu \cos(\nu_\mu t) (\sigma_{34} + \sigma_{43})}_{\text{microwave}}. \end{aligned} \quad (1)$$

The classical fields carry their full time dependence; the cavity coupling carries both a and $a^\dagger$. Nothing has been dropped.

2 Step 1: the rotating frame

Transform with $\tilde{H} = U H U^\dagger + i \dot{U} U^\dagger$, choosing

$$U(t) = \exp \left\{ i t \left[\omega_L a^\dagger a + \omega_L \sigma_{22} + (\omega_L + \nu_c) \sigma_{33} + (\omega_L + \nu_c + \nu_\mu) \sigma_{44} \right] \right\}, \quad (2)$$

i.e. each level is rotated at the *sum of the drive frequencies needed to reach it* (ω_L is a reference near the cavity resonance). This is an exact change of picture. Its whole effect is on the diagonal:

level	rotated at	lab energy	rotating-frame energy
$ 1\rangle$	0	0	0
$ 2\rangle$	ω_L	$2\pi \times 384$ THz	$\Delta \equiv \omega_2 - \omega_L$ ($\sim$ GHz)
$ 3\rangle$	$\omega_L + \nu_c$	$2\pi \times 1009$ THz	$\delta \equiv \omega_3 - \omega_L - \nu_c$ ($\sim$ MHz)
$ 4\rangle$	$\omega_L + \nu_c + \nu_\mu$	$\omega_3 + 2\pi \times 10$ GHz	$\delta + \Delta_\mu, \quad \Delta_\mu \equiv (\omega_4 - \omega_3) - \nu_\mu$

This answers the obvious objection. $|3\rangle$ and $|4\rangle$ are *not* degenerate — they are 10 GHz apart. But the frame rotates $|4\rangle$ faster than $|3\rangle$ by exactly ν_μ , and ν_μ is chosen close to that splitting. The 10 GHz is subtracted off; what remains on the diagonal is only the residual *microwave detuning* Δ_μ , of order MHz. Setting $\Delta_\mu = 0$ makes the two entries equal — and that is a statement about the *drive*, not about the atom.

3 Step 2: the rotating-wave approximation

In the new frame the cavity term $a^\dagger \sigma_{21}$ picks up $e^{2i\omega_L t}$, and $\cos(\nu t) \sigma_{jk}$ splits into a static piece and one at $e^{2i\nu t}$. Dropping the fast pieces requires

$$g_0, \Omega_c, \Omega_\mu \ll \omega_L, \nu_c, \nu_\mu, \quad (\text{A1})$$

which is spectacularly well satisfied (MHz against THz, and MHz against the 10 GHz microwave). The result is time-independent and still exact to this order:

$$\tilde{H} = \omega_c a^\dagger a + \begin{pmatrix} 0 & g_0 a^\dagger & 0 & 0 \\ g_0 a & \Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \omega_c \equiv \omega_{\text{cav}} - \omega_L. \quad (3)$$

Note the matrix is tridiagonal: each field connects only neighbouring levels.

4 Step 3: eliminate the intermediate state

Detune far from the lossy state $|2\rangle$,

$$|\Delta| \gg \Omega_c, g_0 \sqrt{\langle a^\dagger a \rangle}, \Gamma_e, \quad (\text{A2})$$

so its amplitude follows adiabatically: $c_2 \simeq -(g_0 a c_1 + \frac{\Omega_c}{2} c_3)/\Delta$. Substituting into Eq. (3) leaves, in $\{|1\rangle, |3\rangle, |4\rangle\}$,

$$\begin{pmatrix} -\frac{g_0^2 a^\dagger a}{\Delta} & -\lambda a^\dagger & 0 \\ -\lambda a & \delta - \frac{\Omega_c^2}{4\Delta} & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \boxed{\lambda = \frac{g_0 \Omega_c}{2\Delta}} \quad (4)$$

The off-diagonal λa is the two-photon Raman coupling we want. The diagonal entries are light shifts: $-\Omega_c^2/4\Delta$ on $|3\rangle$, absorbed by redefining δ , and a photon-number-dependent shift on $|1\rangle$, negligible for $g_0^2 \langle a^\dagger a \rangle \ll |\Delta| \omega_c$. The overall sign of λ is a phase convention and we drop it.

Where the dominant systematic comes from. $\lambda \propto \Omega_c$ and the light shift $\propto \Omega_c^2$ both track the control-laser intensity, and the shift adds *directly* to δ — the same channel the measurand uses. Control-laser intensity noise is therefore structurally indistinguishable from signal.

5 Step 4: dress the Rydberg pair

The lower-right block of Eq. (4) is a two-level problem on its own:

$$H_\mu = \begin{pmatrix} \delta & \Omega_\mu/2 \\ \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix} = \left(\delta + \frac{\Delta_\mu}{2}\right) \mathbf{1} + \frac{1}{2}(\Omega_\mu \sigma_x + \Delta_\mu \sigma_z). \quad (5)$$

Diagonalise with the rotation $R(\theta)$ defined by

$$\tan \theta = \frac{\Omega_\mu}{\Delta_\mu}, \quad |+\rangle = \sin \frac{\theta}{2} |3\rangle + \cos \frac{\theta}{2} |4\rangle, \quad |-\rangle = \cos \frac{\theta}{2} |3\rangle - \sin \frac{\theta}{2} |4\rangle, \quad (6)$$

giving the *generalised Rabi* splitting

$$\boxed{\omega_\pm = \delta + \frac{\Delta_\mu}{2} \pm \frac{W}{2}, \quad W = \sqrt{\Omega_\mu^2 + \Delta_\mu^2}} \quad (7)$$

Now rotate the rest of Eq. (4). The Raman coupling connects $|1\rangle$ to the *bare* $|3\rangle = \sin \frac{\theta}{2} |+\rangle + \cos \frac{\theta}{2} |-\rangle$, so it splits between the branches in that ratio. In the basis $(|g\rangle, |+\rangle, |-\rangle)$:

$$\boxed{H^{(3)} = \begin{pmatrix} 0 & g_+ a^\dagger & g_- a^\dagger \\ g_+ a & \omega_+ & 0 \\ g_- a & 0 & \omega_- \end{pmatrix}, \quad g_+ = \lambda \sin \frac{\theta}{2}, \quad g_- = \lambda \cos \frac{\theta}{2}} \quad (8)$$

There is no $|+\rangle \leftrightarrow |-\rangle$ entry: that transition is the microwave one and carries no cavity matrix element.

The two limits, and why they are the whole story

	Resonant: $\Delta_\mu = 0$	Far detuned: $ \Delta_\mu \gg \Omega_\mu$
mixing angle	$\theta = \pi/2$	$\theta \rightarrow 0$
branches	$ \pm\rangle = (3\rangle \pm 4\rangle)/\sqrt{2}$	$ -\rangle \rightarrow 3\rangle, +\rangle \rightarrow 4\rangle$
splitting W	Ω_μ (linear in E_μ)	$ \Delta_\mu + \Omega_\mu^2/2 \Delta_\mu $
couplings	$g_\pm = \lambda/\sqrt{2}$ (equal)	$g_- \rightarrow \lambda, g_+ \rightarrow 0$ (one branch)
shift of the coupled level	$\pm\Omega_\mu/2$, linear	$-\Omega_\mu^2/4\Delta_\mu$, quadratic
this is	Autler–Townes	AC Stark
used by	this proposal	Ding <i>et al.</i> , Wang <i>et al.</i>

One mixing angle therefore separates the two ways of doing Rydberg electrometry. On resonance the measurand enters as a level *splitting*, linear in E_μ with the exact dipole scale factor, at the price of two branches that both couple. Far detuned it enters as a *shift*, quadratic in E_μ with a polarizability-dependent factor, but the model is genuinely two-level. There is no setting that gives both.

6 Step 5: many atoms, and the Dicke form

Take N atoms in $|g\rangle$ and work at low excitation, $n_+ + n_- \ll N$ (A3). Holstein–Primakoff replaces each collective excitation by a boson $b_\pm$, and the couplings add constructively, $\lambda \rightarrow \lambda\sqrt{N}$. Write $G_\pm = \sqrt{N} g_\pm$. Equation (8) then gives

$$H = \omega_c a^\dagger a + \sum_{s=\pm} \left[\omega_s b_s^\dagger b_s + G_s (a^\dagger b_s + a b_s^\dagger) \right], \quad (9)$$

which conserves excitation number: starting from $|g, 0\rangle$ nothing ever happens, since it is an exact eigenstate. A Dicke model requires the counter-rotating terms $a^\dagger b_s^\dagger + a b_s$ as well, supplied by a second drive tone (or one modulated tone). With both,

$$\boxed{H = \omega_c a^\dagger a + \sum_{s=\pm} \left[\omega_s b_s^\dagger b_s + G_s (a + a^\dagger)(b_s + b_s^\dagger) \right]} \quad (10)$$

7 Step 6: the threshold

In quadratures $x_j = (o_j + o_j^\dagger)/\sqrt{2}$ the potential energy is $\frac{1}{2}\mathbf{x}^\top V\mathbf{x}$,

$$V = \begin{pmatrix} \omega_c & 2G_+ & 2G_- \\ 2G_+ & \omega_+ & 0 \\ 2G_- & 0 & \omega_- \end{pmatrix}. \quad (11)$$

The normal phase (all atoms in $|g\rangle$, cavity dark) is stable while V is positive definite, and loses stability at $\det V = 0$:

$$\boxed{\omega_c \omega_+ \omega_- = 4(G_+^2 \omega_- + G_-^2 \omega_+)} \quad (12)$$

Positive definiteness also demands $\omega_- > 0$; if the lower branch falls below $|g\rangle$ the normal phase fails for a mundane reason — a two-photon resonance — rather than by symmetry breaking. With $G = \lambda\sqrt{N}$ the two limits read

$$\Delta_\mu = 0 : \quad \omega_c \omega_+ \omega_- = 2G^2(\omega_+ + \omega_-) = 4G^2\delta, \quad (13)$$

$$|\Delta_\mu| \gg \Omega_\mu : \quad \omega_c \omega_- = 4G^2, \quad (14)$$

the second being the textbook two-level Dicke threshold $G^c = \frac{1}{2}\sqrt{\omega_c \omega_-}$ for a spin of splitting ω_- . *This* is the effective two-level model — and it is legitimate only in the far-detuned limit, where the measurand has become quadratic.

8 Summary

Step	Condition	Effect
Rotating frame	exact	subtracts the drive frequencies; the 10 GHz $ 3\rangle$ – $ 4\rangle$ splitting becomes Δ_μ
RWA	$g_0, \Omega_c, \Omega_\mu \ll \omega_L, \nu_c, \nu_\mu$	removes all time dependence; Eq. (3)
Eliminate $ 2\rangle$	$ \Delta \gg \Omega_c, g_0\sqrt{n}, \Gamma_e$	$\lambda = g_0\Omega_c/2\Delta$, plus light shifts (the dominant systematic)
Dress $ 3\rangle, 4\rangle$	exact	$\omega_\pm = \delta + \Delta_\mu/2 \pm W/2$, couplings $\lambda \sin \frac{\theta}{2}$, $\lambda \cos \frac{\theta}{2}$
Holstein–Primakoff	$n_+ + n_- \ll N$	$\lambda \rightarrow G = \lambda\sqrt{N}$
Counter-rotating	second drive tone	without it $ g, 0\rangle$ is an eigenstate and nothing radiates
Two-level limit	$ \Delta_\mu \gg \Omega_\mu$	one branch decouples; measurand becomes quadratic

Only the last step is optional, and it is the one that decides what kind of sensor this is.